

16.

Ans: B

Initially, since the oil droplet drops with a reduced acceleration, it means the electric force due to the two metal plates is acting in the opposite direction.

$$W - F_E = ma$$

Subsequently, when the polarity is reversed, we have

$$W + F_E = 7ma$$

Thus by subtracting the first expression from the second

$$2F_E = 6ma$$

$$2 \frac{qV}{x} = 6ma$$

$$q = \frac{3max}{V}$$

17.

Ans: A